Virgo SLM V1.0 Report Blueprint

Building a 120M Parameter Transformer Language Model from Scratch

A Self-Learning Engineering Project

This blueprint is intended for the final public report. The report should read like the work of a curious first-year undergraduate documenting an engineering journey rather than an industrial research paper. Focus on what was built, what was learned, what failed, and how problems were solved.

Cover Page

Title, subtitle, author (Punit Kumar Kashyap), First-Year Undergraduate, Department of Engineering Physics, IIT Dharwad, July 2026.

Chapter 1 – Why I Started Virgo

Motivation, curiosity about language models, learning goals, project philosophy.

Chapter 2 – Learning Journey

Timeline from Python to Transformers, tokenizer, datasets, training, fine-tuning, inference.

Chapter 3 – End-to-End Pipeline

Large flowchart: Raw Data → Cleaning → Tokenizer → Tokenization → Pretraining → Virgo Base → Instruction Tuning → Virgo Chat.

Chapter 4 – Dataset

Data sources, preprocessing, balancing, cleaning pipeline, pie charts and token distribution.

Chapter 5 – Tokenizer

Byte-Level BPE, vocabulary, special tokens, BPE example diagrams.

Chapter 6 – Transformer Architecture

Token embedding, RoPE, attention, decoder block, FFN, residuals, layer norm, architecture diagrams.

Chapter 7 – Training

Hardware, hyperparameters, training loop, checkpoints, loss curves, learning-rate schedule.

Chapter 8 – Challenges

Dataset bugs, tokenizer issues, CUDA OOM, failed runs, debugging lessons.

Chapter 9 – Demonstrations

Example prompts and responses for coding, writing, reasoning, summarization. Avoid publishing benchmark scores.

Chapter 10 – Lessons Learned

What the project taught about AI engineering and research.

Chapter 11 – Future Work

Reasoning, larger models, multimodal, better evaluation, improved datasets.

Appendix

Repository structure, configs, tokenizer stats, model configuration, acknowledgements.

Recommended diagrams (20–30): overall pipeline, dataset flow, tokenizer workflow, BPE example, transformer block, attention, RoPE, FFN, residual connection, training loop, checkpoint system, inference pipeline, web app architecture, folder structure, token distribution, loss curves, learning roadmap, model configuration, preprocessing pipeline, instruction tuning workflow.

Recommended Release Name: Virgo SLM V1.0

Subtitle: Building a 120M Parameter Transformer Language Model from Scratch

Report Type: A Self-Learning Engineering Project